

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to define the functional and non-functional requirements of the **Insurance Broking System with Machine Learning Analytics**. This system is designed to help insurance brokers manage leads, customers, policies, and generate predictive insights such as lead conversion and churn probability.

1.2 Scope

The system provides a centralized platform that includes:

- Customer and lead management
- Policy tracking and management
- Payment and commission tracking
- Predictive analytics using machine learning
- Dashboard and reporting features

The system does not create insurance products but facilitates the management and analytics of existing policies.

1.3 Intended Users

- Insurance Brokers
 - Sales Agents
 - System Administrators
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1.4 Assumptions and Dependencies

- Users have basic knowledge of insurance processes
 - System requires internet connectivity
 - Data is stored in a centralized cloud database
 - Machine learning models depend on historical/synthetic data
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2. Overall Description

2.1 Product Perspective

The system is a web-based centralized platform that replaces manual processes and fragmented data systems. It integrates operations with machine learning analytics for better decision-making.

2.2 Product Functions

- Lead and customer management
 - Policy lifecycle management
 - Payment and commission tracking
 - Document management
 - Dashboard and reporting
 - Machine learning-based predictions
 - Secure authentication and access control
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3. Functional Requirements

3.1 Customer and Lead Management

- Add, update, delete customer and lead data
 - Convert leads into customers
 - Track lead status (open, converted, rejected)
 - Store customer profiles and history
 - Search and filter records
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3.2 Policy Management

- Create and manage insurance policies
 - Store policy details (number, premium, duration, status)
 - Support multiple policy types (life, health, motor, etc.)
 - Update and track policy lifecycle
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3.3 Payment and Commission Management

- Record payments (amount, mode, date)
 - Track commission for agents
 - Maintain financial records
 - Generate reports
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3.4 Predictive Churn Analysis

- Identify customers likely to discontinue policies
- Assign churn risk score
- Categorize customers (High / Medium / Low risk)

- Display results on dashboard
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3.5 Lead Conversion Scoring

- Predict probability of lead conversion
 - Categorize leads (Hot, Warm, Cold)
 - Prioritize leads automatically
 - Improve sales efficiency
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3.6 Dashboard and Reporting

- Display key metrics (customers, policies, revenue)
 - Provide real-time updates
 - Generate reports (PDF/CSV)
 - Visualize data using charts and summaries
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3.7 Document and Endorsement Management

- Upload and store documents securely
 - Link documents to policies/customers
 - Track policy endorsements and updates
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3.8 User and Role Management

- User authentication (login system)
 - Role-based access control
 - Activity logging
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4. Non-Functional Requirements

4.1 Performance Requirements

- Fast response time
 - Support multiple users
 - Efficient handling of large datasets
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4.2 Security Requirements

- Secure login authentication
- Data encryption

- Role-based access control
 - Activity logs
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4.3 Usability Requirements

- User-friendly interface
 - Easy navigation
 - Accessible via web browsers
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4.4 Reliability and Scalability

- High system availability
 - Data consistency and integrity
 - Scalable for future growth
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5. System Requirements

5.1 Database Requirements

- PostgreSQL (via Supabase)
 - Relational database structure
 - Support for real-time updates
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5.2 Software Requirements

- Frontend: FlutterFlow
 - Backend: Supabase
 - Machine Learning: Python (Scikit-learn)
 - Browser-based access
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5.3 Hardware Requirements

- Minimum 4GB RAM
 - Internet connection
 - Web browser
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6. Constraints

- No insurance product creation
- No claim processing

- Works only online
 - Limited real-world data (uses synthetic data)
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7. Conclusion

This SRS defines the complete requirements of an intelligent insurance broking system that integrates operational management with machine learning analytics. The system aims to improve efficiency, decision-making, and customer retention.